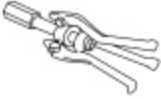
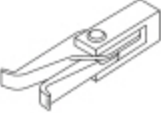
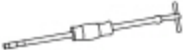


Transfer Case — Electronic Shift

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

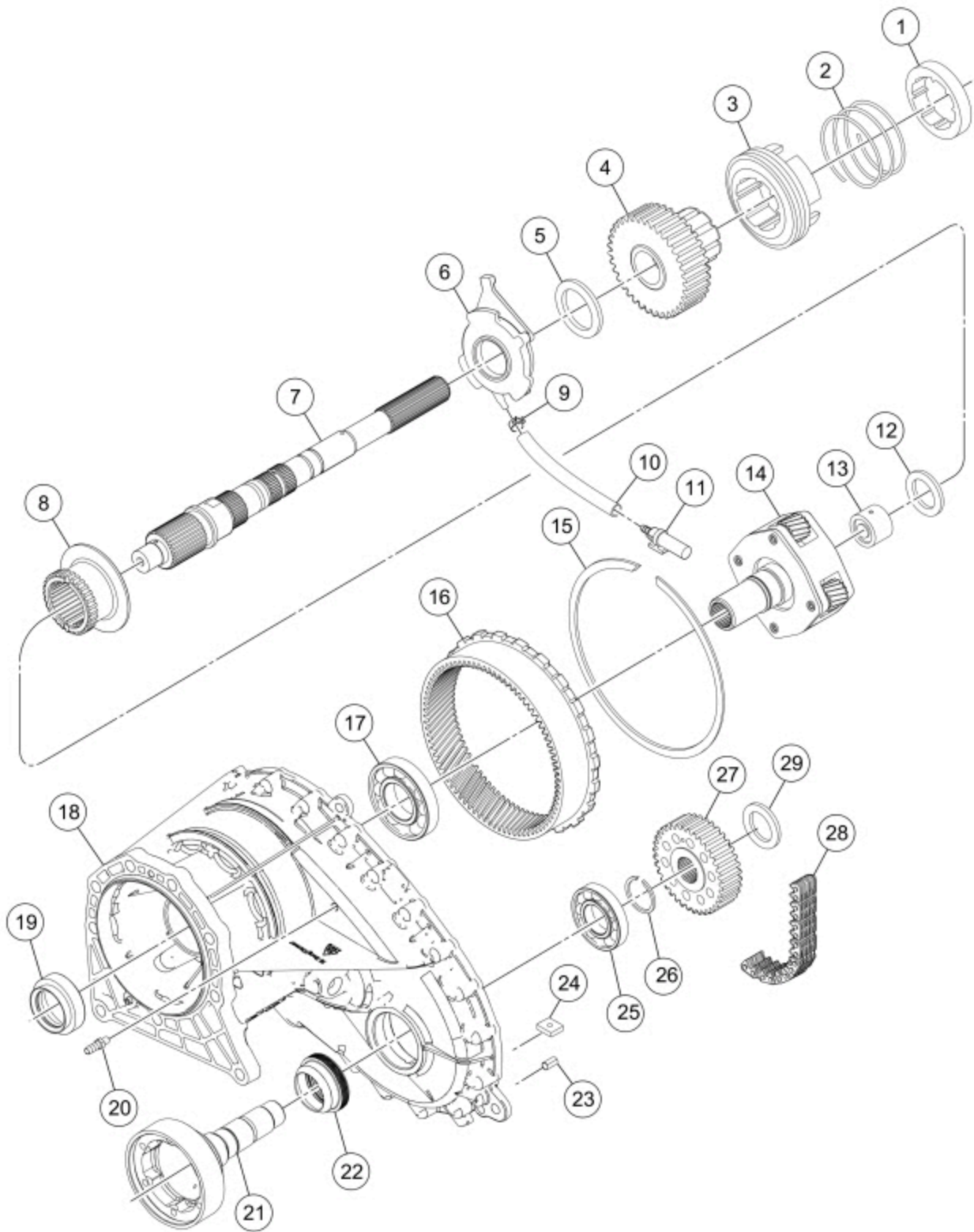
 ST1255-A	Handle 205-153 (T80T-4000-W)
 ST1186-A	Holding Fixture, Transmission 307-003 (T57L-500-B)
 ST1471-A	Installer, Bearing Cup 308-017 (T73T-4222-B)
 ST1789-A	Installer, Input Shaft Bearing 308-085 (T83T-7025-C)
 ST2305-A	Installer, Input Shaft Oil Seal 308-186 (T90T-7127-B)
 ST2261-A	Installer, Input Shaft Oil Seal 308-249 (T96T-7127-A)
 ST1788-A	Installer, Mainshaft Bearing 308-060 (T77J-7025-K)

 ST1200-A	Remover, Bearing Cup 308-047 (T77F-1102-A)
 ST1362-A	Remover, Stator Bearing 307-318 (T94P-77001-KH)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)

Material

Item	Specification
Ultra Silicone Sealant TA-29	—

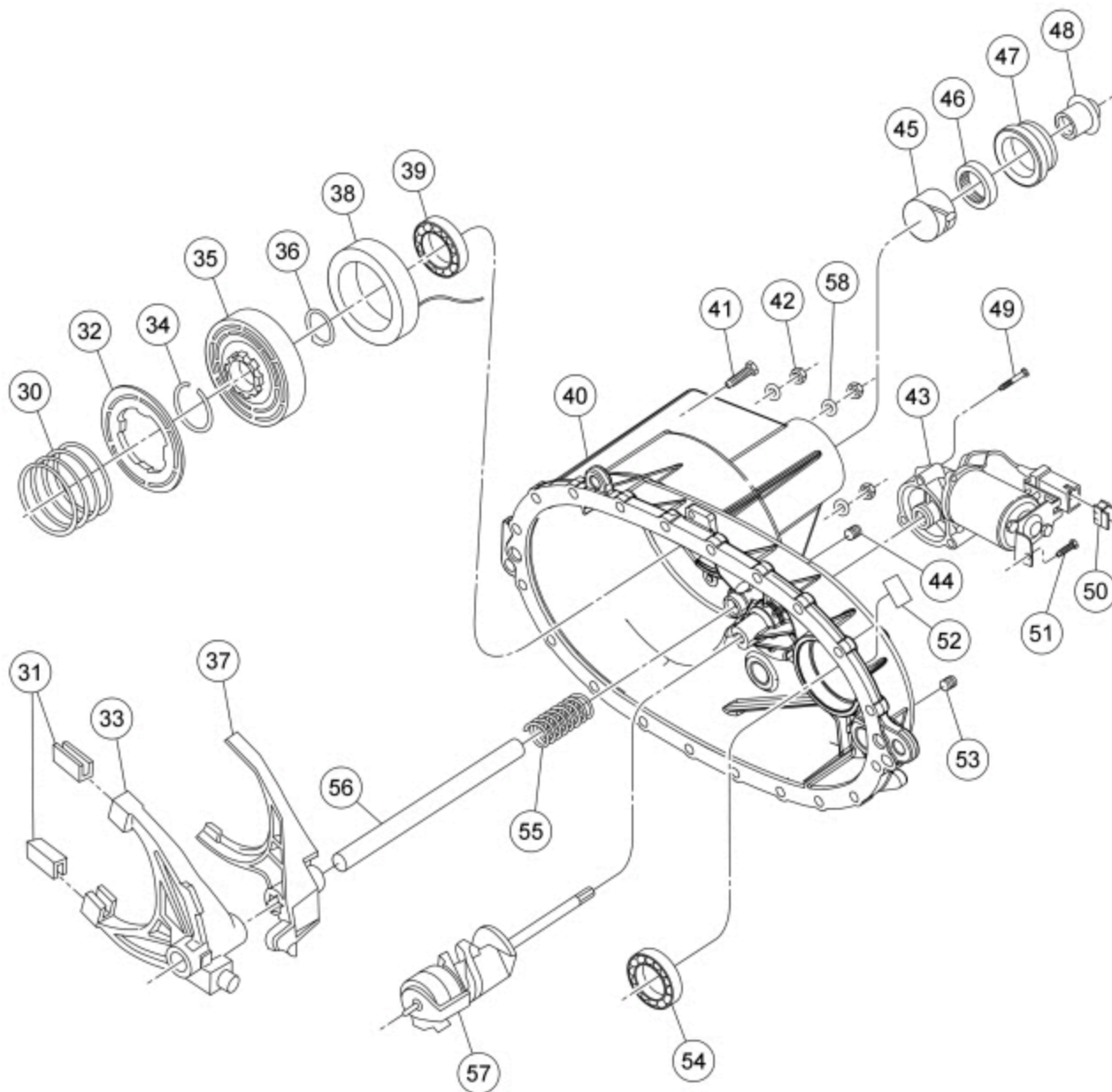
Transfer Case Exploded View — Electronic Shift



N0171188

Item	Part Number	Description
1	7D164 7D164	Lockup hub
2	7D126 7D126	Spring
3	7106 7106	Lockup collar
4	7177 7177	Drive sprocket
5	7Z112 7Z112	Thrust washer
6	7A149 7A149	Oil pump assembly
7	7061 7061	Output shaft

8	7100 7100	Reduction hub
9	—	Hose clamp
10	9324 9324	Hose
11	7A098 7A098	Lube pickup and filter
12	7D221 7D221	Thrust bearing
13	71207 7120	Needle bearing
14	7A398 7A398	Front planet carrier
15	7064 7064	Snap ring
16	7A153 7A153	Ring gear
17	7025 7025	Bearing
18	7005 7005	Case
19	7B215 7B215	Input flange seal
20	7034 7034	Vent tube
21	7061 7061	Front output shaft
22	7B215 7B215	Front output shaft seal
23	—	Dowel (part of 7005)
24	7E290 7E290	Magnet
25	7025 7025	Bearing
26	79177 7917	Snap ring
27	71777 7177	Driven sprocket
28	7A029 7A029	Drive chain assembly
29	7N063 7N063	Driven sprocket spacer (If equipped)



N0130854

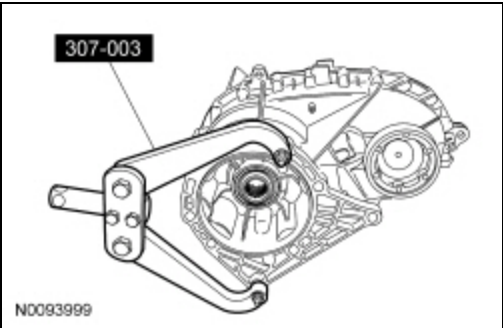
Item	Part Number	Description
30	7F1357F135	Spring
31	7C4307C430	Shift fork pads
32	7H1507H150	Armature
33	72897289	Reduction shift fork assembly
34	79177917	Snap ring
35	7G3627G362	Hub and coil housing
36	79177917	Snap ring

37	7289 7289	Lockup fork
38	7G361 7G361	Transfer case field coil
39	7025 7025	Bearing
40	7005 7005	Case
41	7A443 7A443	Bolt
42	N62048C N62048C -S2-S2	Nut
43	7G360 7G360	Transfer case shift motor
44	7A010 7A010	Case plug
45	—	Bushing (part of 7005)
46	—	Sealing ring (part of 7005)
47	7B215 7B215	Rear output shaft seal
48	7F293 7F293	Shipping plug
49	N800670 N800670 -S-S	Bolt
50	—	Wire connector spacer (part of 7G360)
51	7A443 7A443	Bolt
52	—	Bar code label (part of 7005)
53	7A010 7A010	Case plug
54	7025 7025	Bearing
55	7219 7219	Lockup fork spring
56	7240 7240	Shift rail
57	7F063 7F063	Shift cam assembly
58	7B368 7B368	Coil nut washer

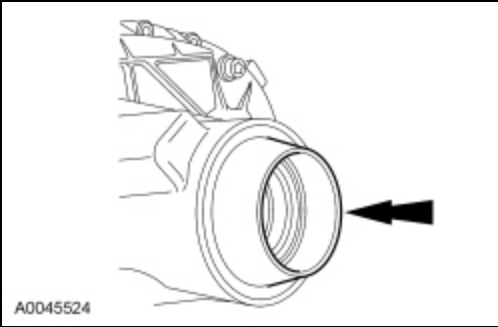
Disassembly

NOTE: *Discard all seals after removing them.*

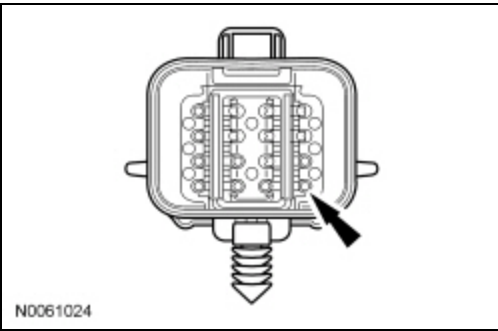
1. Clean the transfer case with solvent and dry with compressed air.
2. Using the Transmission Holding Fixture, secure the transfer case to the bench.



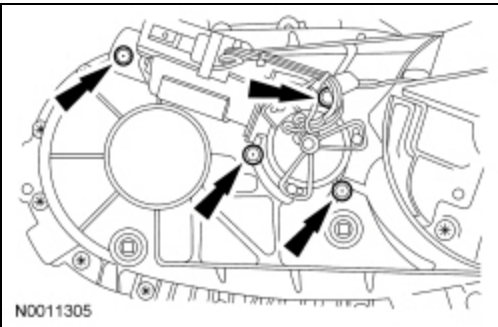
3. Remove the rear output shaft seal.



4. Remove the inner connector retainer. Press the release tab and remove the coil wire from the electric shift motor electrical connector.

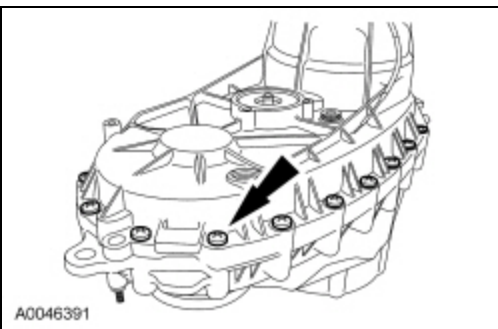


5. Remove the 3 transfer case shift motor bolts, bracket bolt and loosen the 2 bracket nuts.

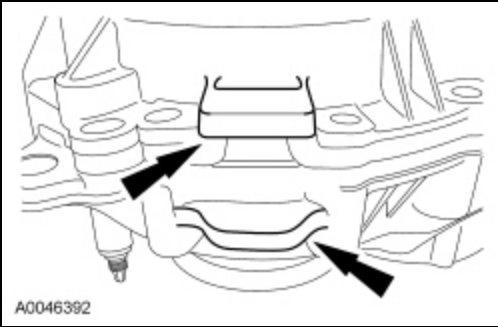


6. **NOTE:** *The transfer case bolts are self tapping, and it is normal to find metal shavings while removing the cover.*

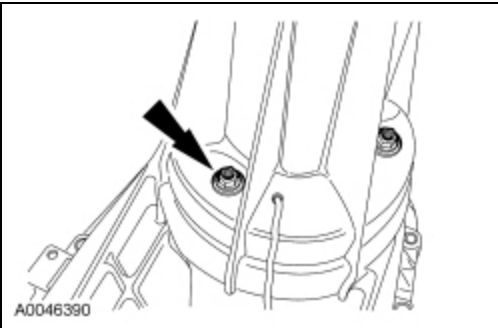
Remove the 20 transfer case bolts.



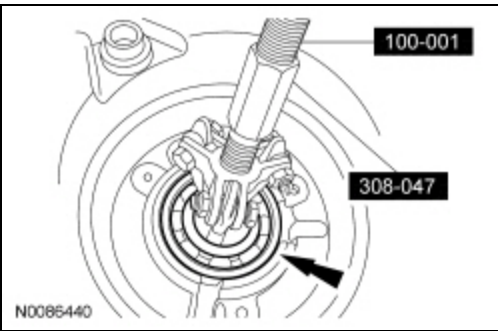
7. Using the pry bosses, separate the transfer case covers.



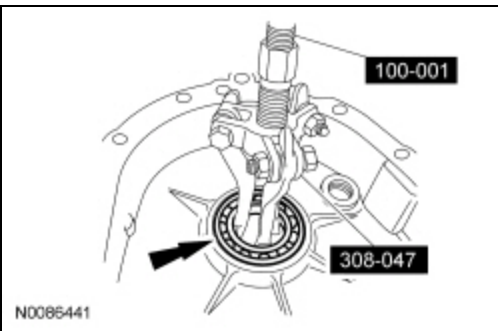
8. Remove the 3 transfer case field coil nuts, washers and transfer case field coil.



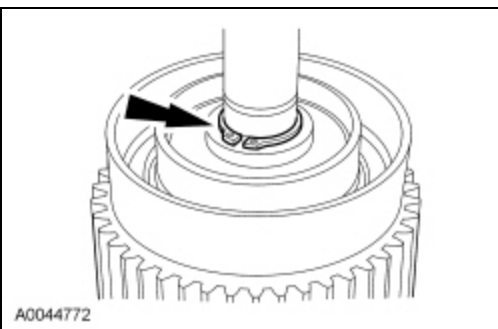
9. Using the Bearing Cup Remover with the Slide Hammer, remove the rear output shaft bearing.



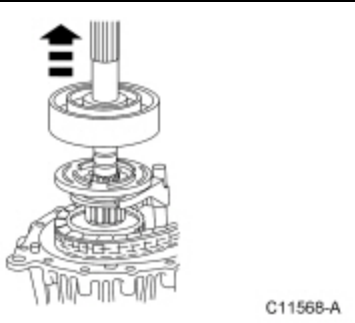
10. Using the Bearing Cup Remover with the Slide Hammer, remove the front output shaft bearing.



11. Remove the shift collar hub snap ring.

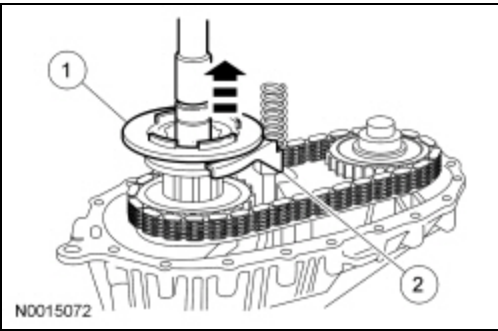


12. Remove the coil housing.



13. Remove the shift fork.

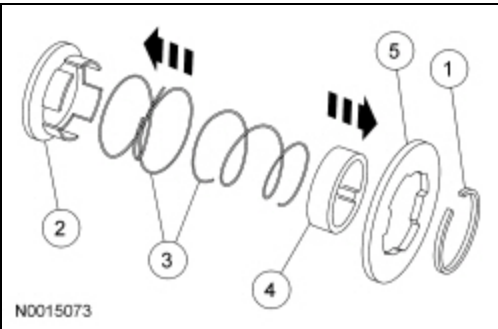
1. Remove the lockup hub.
2. Remove the shift fork and spring.



14. **NOTE:** Remove the lockup hub assembly as a single unit. If disassembly is required, proceed with the following.

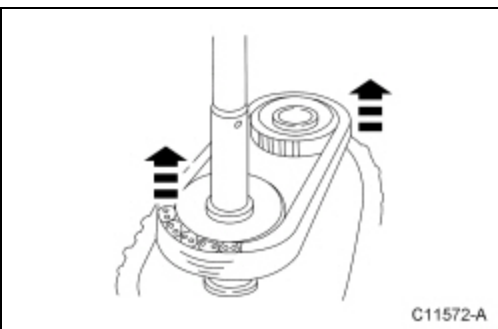
Disassemble the lockup hub assembly.

1. Remove the lockup hub snap ring.
2. Separate the lockup collar.
3. Separate the 2 lockup springs.
4. Separate the lockup hub.
5. Remove the armature.

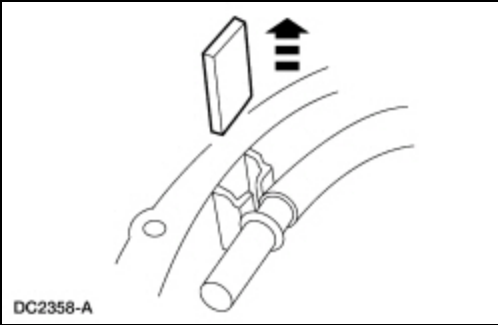


15. **NOTE:** Remove the drive chain and sprockets as an assembly.

Remove the drive chain, sprockets and the thrust washer.

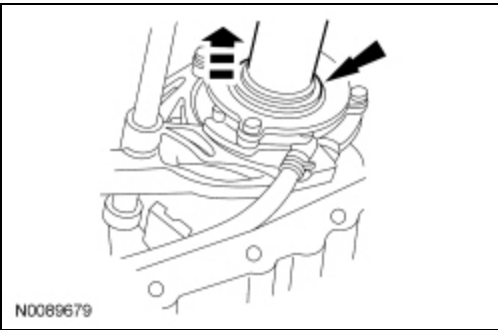


16. Remove the magnet from its slot in the case.

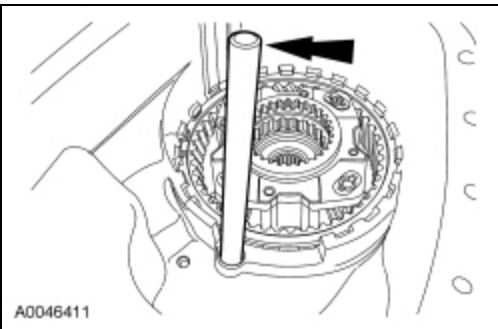


17. **NOTE:** *The pump assembly is not repairable.*

Remove the oil pump and the output shaft as an assembly. Remove the oil pump from the output shaft.

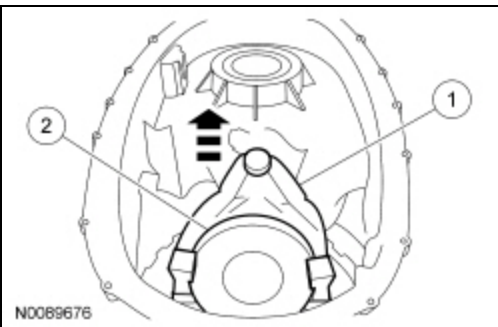


18. Remove the shift rail.



19. Remove the shift fork and the reduction hub.

1. Remove the shift fork.
2. Remove the reduction hub.



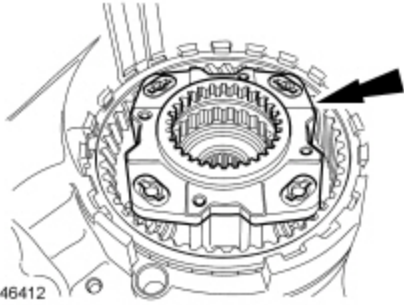
20. **NOTE:** *The electric shift cam is serviced as an assembly. Do not disassemble the shift cam assembly.*

Remove the electric shift cam assembly.



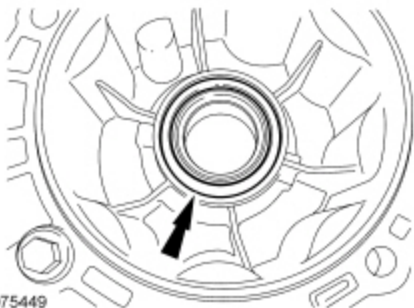
C11703-A

21. Remove the front planet carrier.



A0046412

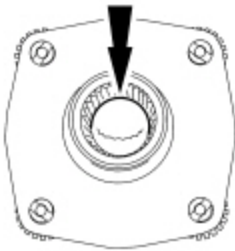
22. Remove and discard the front input shaft seal.



ND075449

23. Remove the thrust bearing.

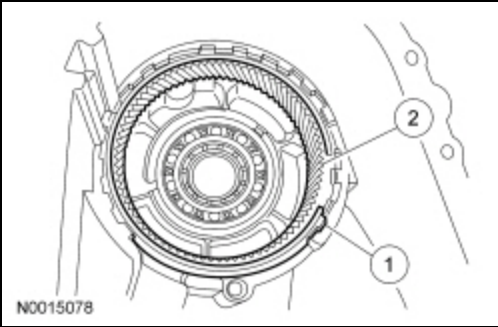
- Remove the needle bearing, if necessary.



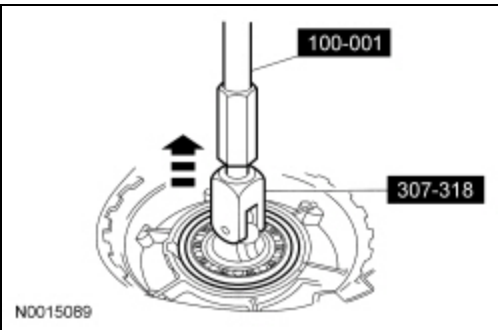
A0044773

24. Remove the ring gear.

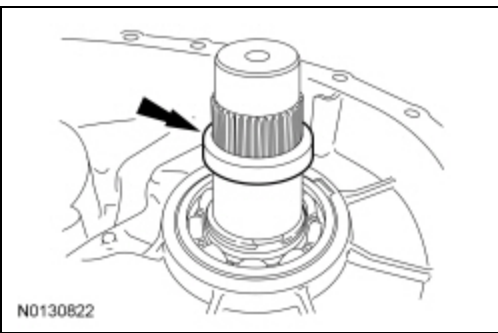
1. Remove the ring gear snap ring.
2. Remove the ring gear.



25. Using the Stator Bearing Remover with the Slide Hammer, remove the front planet carrier bearing.



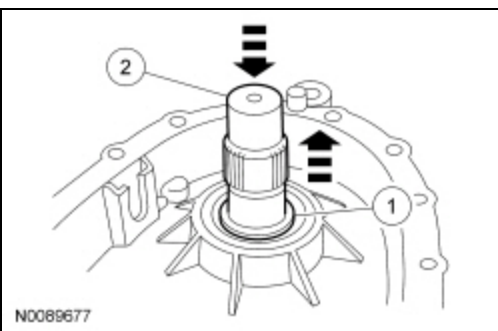
26. If equipped, remove the driven sprocket spacer.



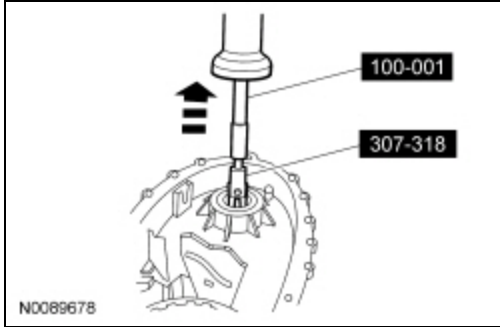
27. **NOTICE: Do not let the front output shaft fall while removing the snap ring or damage to the component may occur.**

Remove the front output shaft.

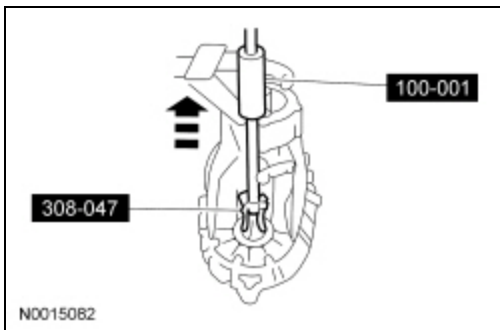
1. Remove the front output shaft snap ring.
2. Remove the front output shaft.



28. Using the Stator Bearing Remover with the Slide Hammer, remove the front output shaft bearing.

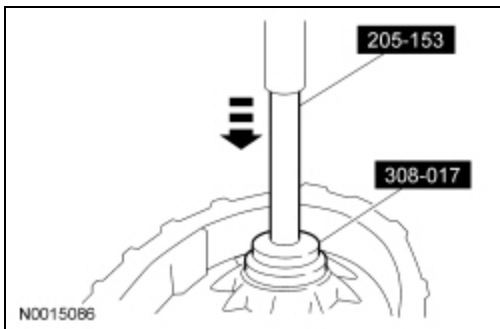


29. Using the Bearing Cup Remover with the Slide Hammer, remove the front output shaft seal.

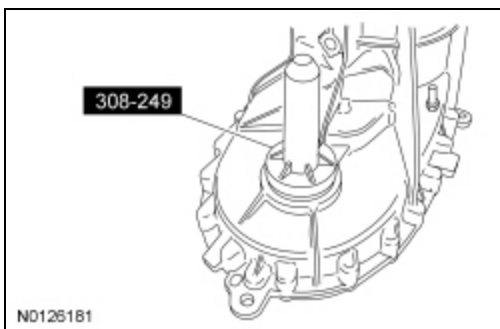


Assembly

1. Using a press and the Bearing Cup Installer with the Handle, install the front output shaft bearing.

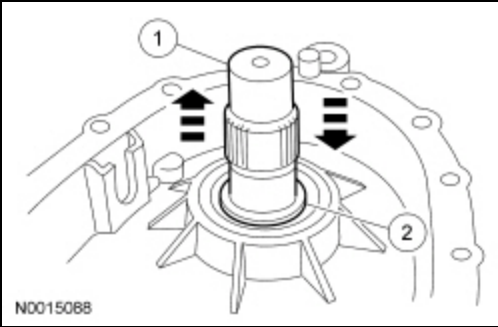


2. Using the Input Shaft Oil Seal Installer, install the front output shaft seal.

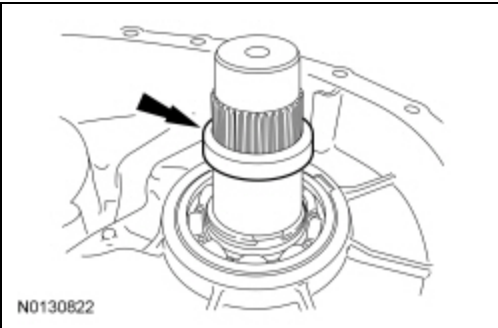


3. Install the front output shaft.

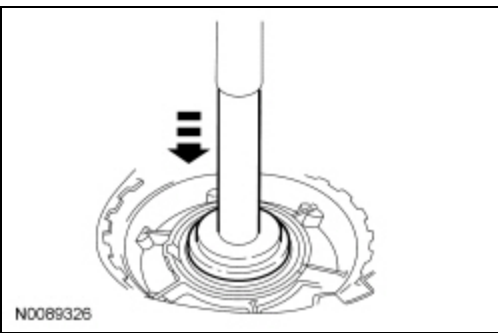
1. Position the front output shaft.
2. Install the front output shaft snap ring.



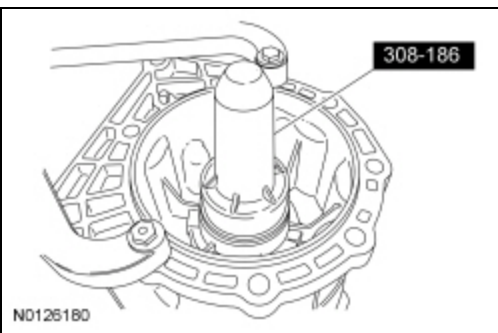
4. If equipped, install the driven sprocket spacer.



5. Install the front planet carrier bearing.

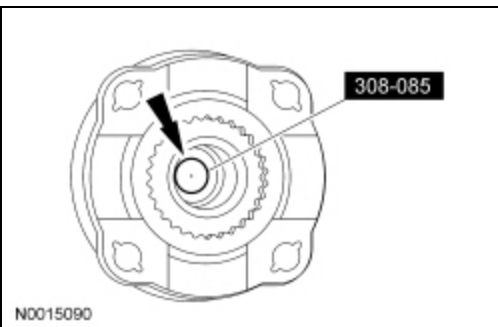


6. Using the Input Shaft Oil Seal Installer, install the front input seal.

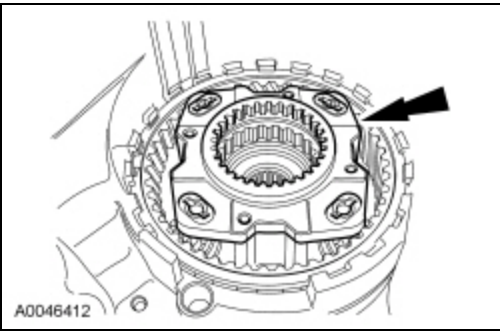


7. Position the ring gear in the case. Install the ring gear snap ring.

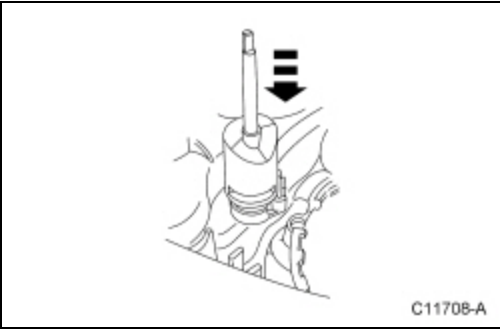
8. If removed, using a press and the Input Shaft Bearing Installer, install the needle bearing into the front planet carrier.



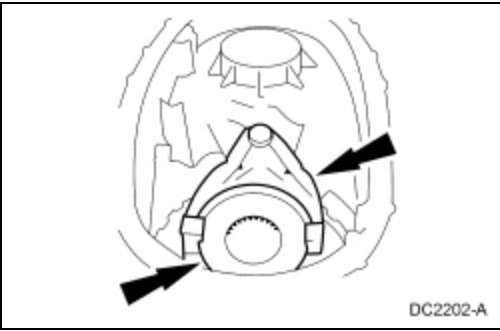
9. Install the front planet carrier and the thrust bearing.



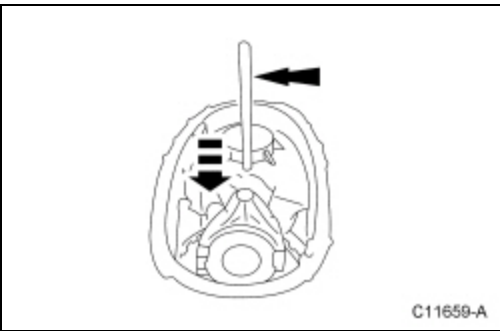
10. Install the electric shift cam assembly.



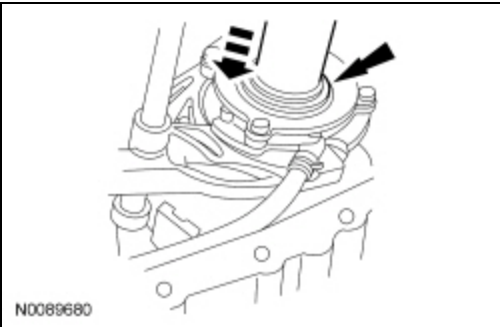
11. Install the shift fork and the reduction hub.



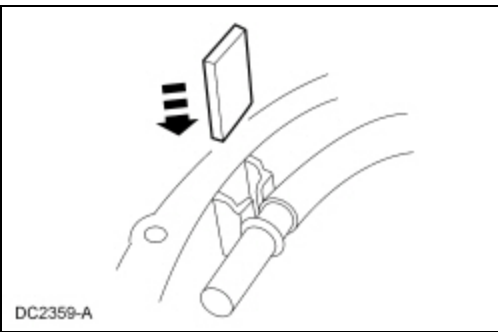
12. Install the shift rail.



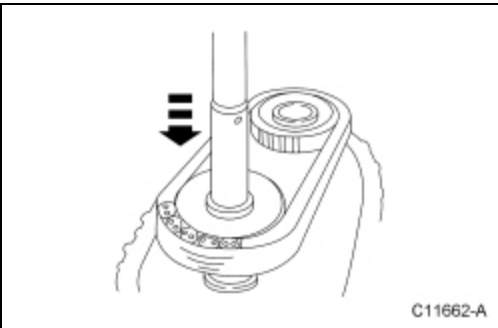
13. Install the oil pump, output shaft and the thrust washer.



14. Install the magnet into its slot in the case.

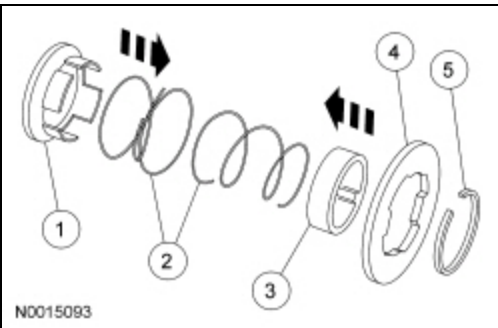


15. Install the drive chain and the sprockets as an assembly.

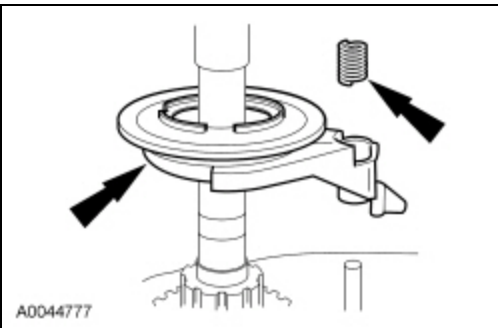


16. Install the lockup hub assembly. If the lockup hub assembly was disassembled, carry out the following steps.

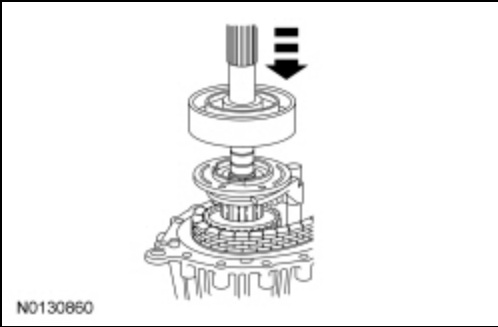
1. Position the lockup collar.
2. Install the lockup collar springs.
3. Install the lockup hub.
4. Install the armature.
5. Install the lockup hub snap ring.



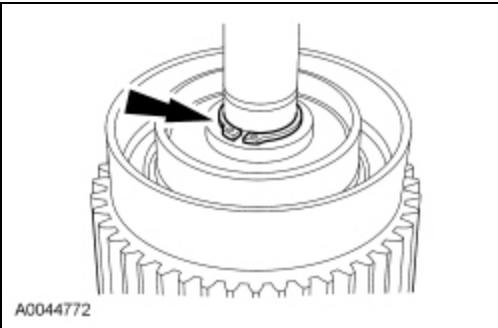
17. Install the lockup hub and the shift fork and spring.



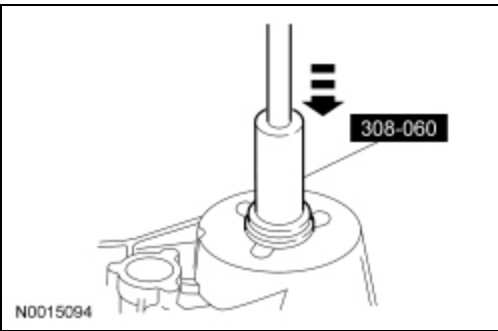
18. Install the coil housing.



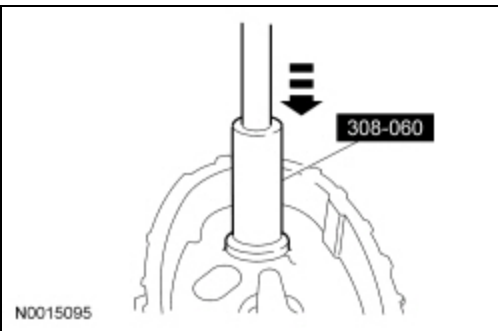
19. Install the snap ring.



20. Using the Mainshaft Bearing Installer, install the rear output shaft bearing.

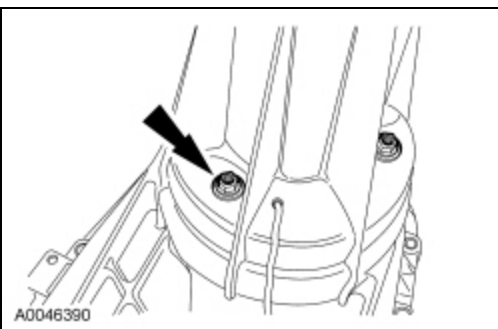


21. Using the Mainshaft Bearing Installer, install the front output shaft bearing.



22. Install the transfer case field coil, washers and the 3 nuts.

- Tighten to 10 Nm (89 lb-in).

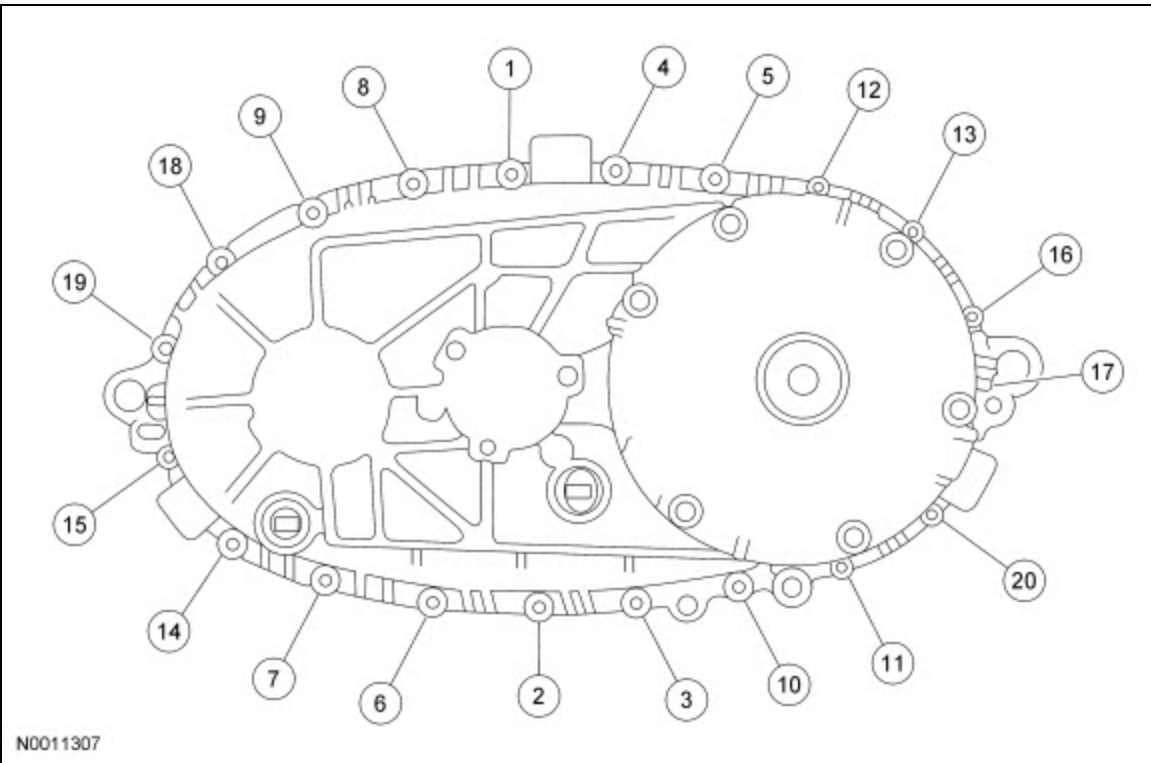


23. **NOTE:** Only apply enough silicone sealant to seal the transfer case halves together.

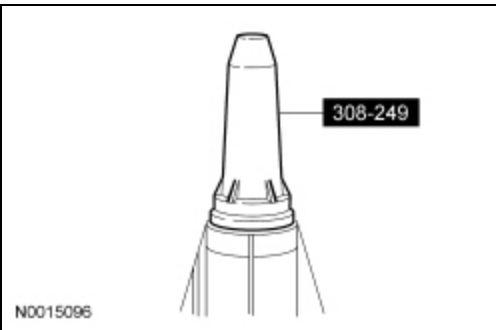
Apply a small bead of silicone sealant to the mating surface of the case and cover.

24. Position the 2 transfer case halves and tighten the 20 bolts in the sequence shown.

- Tighten to 24 Nm (18 lb-ft).



25. Using the Input Shaft Oil Seal Installer, install the rear output shaft seal.

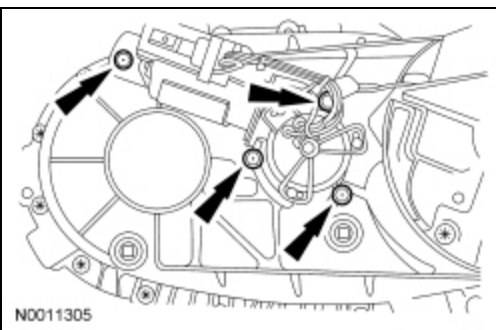


26. Apply a small bead of silicone sealant to the transfer case shift motor mating surface.

NOTICE: The following steps need to be performed in the correct order when installing the transfer case shift motor. Failure to follow the steps in the correct order may result in damage to the component.

27. Install the transfer case shift motor and the 3 bolts.

- Tighten to 10 Nm (89 lb-in).



28. Install the transfer case shift motor bracket bolt.

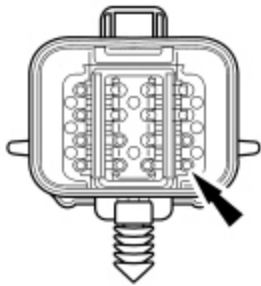
- Tighten to 10 Nm (89 lb-in).

29. Secure the 2 transfer case shift motor bracket nuts.

- Tighten to 3 Nm (27 lb-in).

30. **NOTE:** *After the coil wire is inserted into the connector, gently pull the wire back to verify the pin is locked inside the connector.*

Insert the coil wire pin in the electrical connector and install the inner connector retainer.



31. Remove the transfer case from the Transmission Holding Fixture.

